Project 3: Teen Mental Health: Exploratory Data Analysis Report

Abstract

About The Data

The dataset is taken from a website called Kaggle, an online platform for data science and machine learning. Kaggle has 714,560 datasets available to the public. This dataset is titled “Social Media Impact on Teen Mental Health” with 1,200 observations and 13 columns tracking daily habits and teen traits.

Variable Descriptions

Table 1: Dataset Variables

Variable.Name	Data.Type	Description
age	int	Age of the teen
gender	chr	Gender of teen
daily_social_media_hours	dbl	Average daily hours spent on social media
platform_usage	chr	The main social media platform used
sleep_hours	dbl	Average hours of sleep per night
screen_time_before_sleep	dbl	Hours of screen time before bed
academic_peformance	dbl	Grade Point Average
physical_activity	dbl	Daily hours of physical exercise
social_interaction_level	chr	Level of real-world socialization (low, medium high)
stress_level	int	Stress score from 1-10
anxiety_level	int	Anxiety score from 1-10
addiction_level	int	Social media addiction score from 1-10
depression_label	int	Depression label (0 or 1 indicating whether the teen is classified as depressed)

*Corresponding author

Exploratory Data Analysis

Plot 1 displays the distribution of stress level scores (on a scale from 1-10) of teenagers in the mental health dataset. The results show that stress level is relatively uniform, with the highest frequency count at 4 and the lowest count at 8. Because the distribution is evenly spread, this suggests this sample of teens experiences diverse stress levels, with no trend. For these teens, feeling low stress (1–2), moderate stress (4–6), or extreme stress (9–10) are all nearly equally common. There is no common stress level.

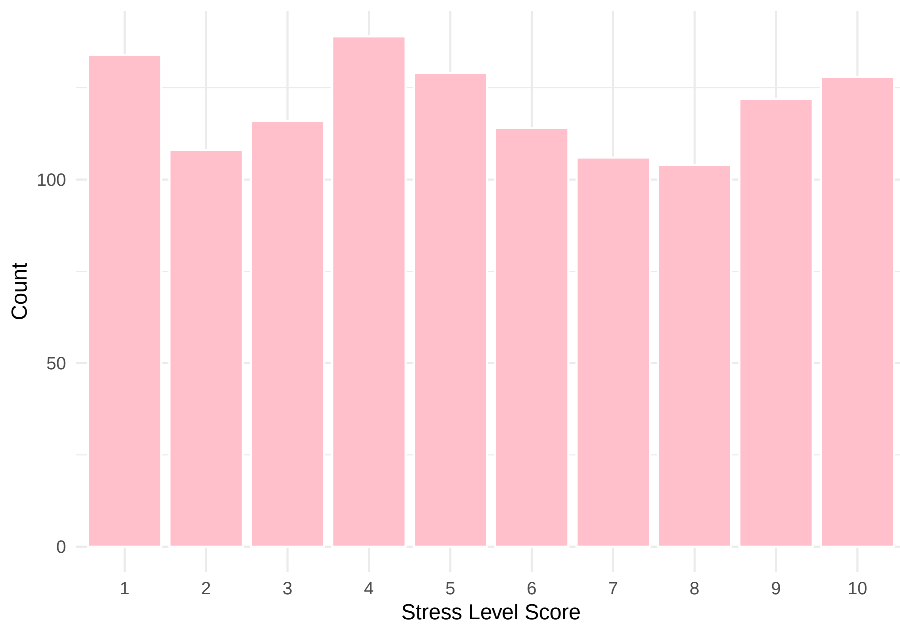

Figure 1: Distribution of Teen Stress Level

Plot 2 illustrates the relationship between hours of screen time before sleep and total hours of sleep per night. Using method = "lm," a linear model was fit to the scatterplot. This line reveals a nearly flat correlation, indicating that there is no significant relationship between screen time before bed and the number of hours slept.

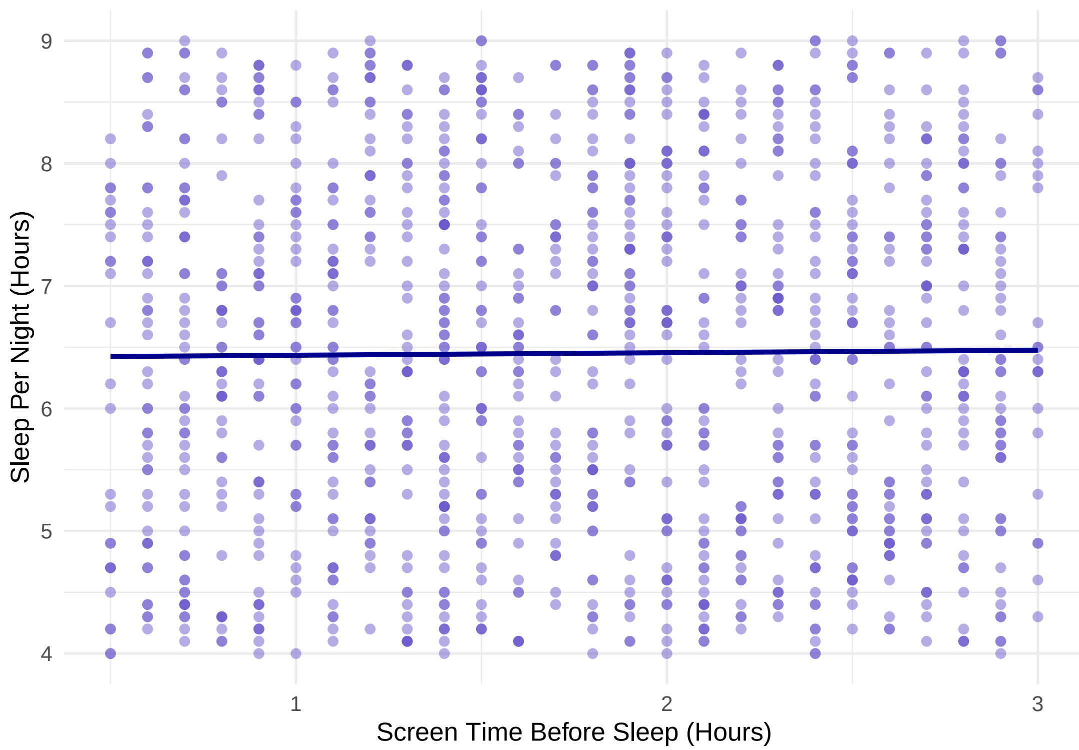

Figure 2: Relationship Between Screen Time Before Sleep and Total Sleep Hours

Plot 3 compares the distribution of anxiety scores (on a scale from 1-10) between teens classified as depressed (depression_label = 1) and those not (depression_label = 0). The boxplots show that teens with depression have a higher median anxiety score. This suggests that there is an association between depression and anxiety levels among the teens in this dataset, with depressed teens tending to have higher anxiety levels. It is important to note the significant difference in group size, with only 31 teens classified as depressed in the dataset. A larger sample size would be helpful to analyze this relationship more robustly.

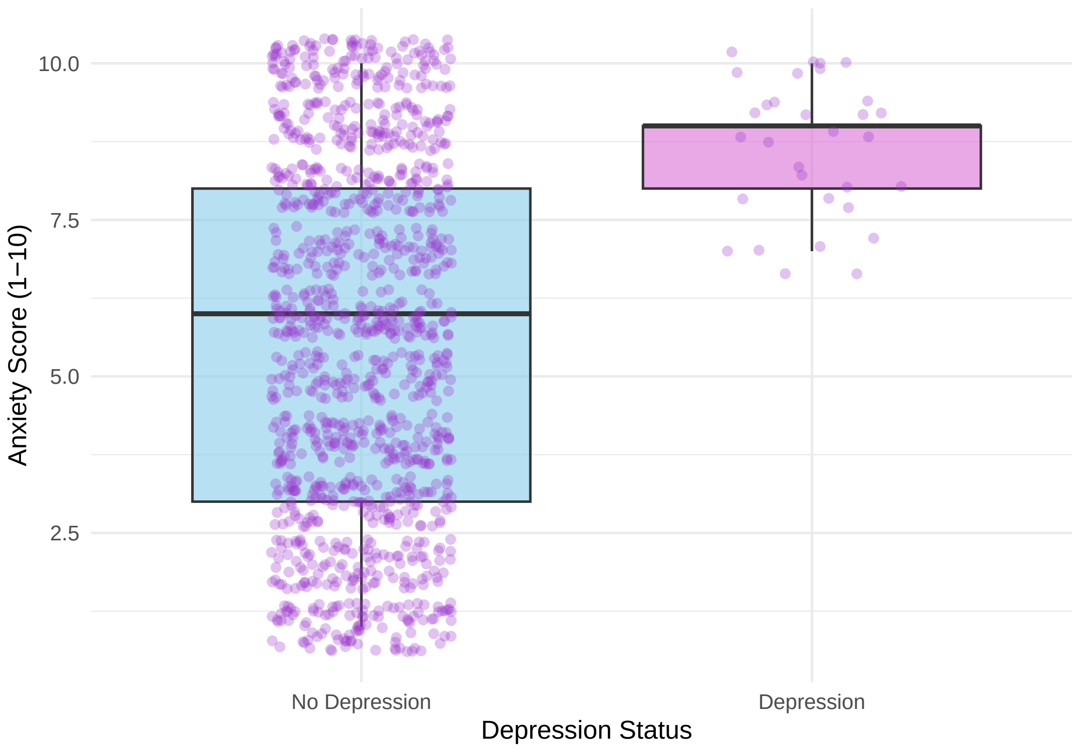

Figure 3: Distribution of Anxiety Scores Among Depression Categories

Summary Statistics

Table 2: Summary Statistics for Key Metrics

Daily Social Media Hours	Daily Hours of Sleep	GPA	Stress Level	Anxiety Level
Min. :1.000	Min. :4.000	Min. :2.00	Min. : 1.000	Min. : 1.000
1st Qu.:2.800	1st Qu.:5.200	1st Qu.:2.50	1st Qu.: 3.000	1st Qu.: 3.000
Median :4.500	Median :6.500	Median :2.99	Median : 5.000	Median : 6.000
Mean :4.537	Mean :6.449	Mean :2.99	Mean : 5.446	Mean : 5.637
3rd Qu.:6.300	3rd Qu.:7.600	3rd Qu.:3.48	3rd Qu.: 8.000	3rd Qu.: 8.000
Max. :8.000	Max. :9.000	Max. :4.00	Max. :10.000	Max. :10.000

References

- Kaggle Dataset: "Social Media Impact on Teen Mental Health" (1,200 Observations). Retrieved June 2026 from https://www.kaggle.com.